Vectors

A *vector* is just a list of numbers. The *dimension* of the vector is the number of numbers in the list. If the lists are written vertically, we call them *column vectors* or *kets*. If the lists are written horizontally, we call them *row vectors* or *bras*. The numbers that make up a vector are often called *entries*. To illustrate, here is a three-dimensional ket and a four-dimensional bra:

$$\begin{bmatrix} 2 \\ 0.5 \\ -3 \end{bmatrix}, \begin{bmatrix} 1 & 0 & -\pi & 23 \end{bmatrix}.$$

The names *bra* and *ket* come from Paul Dirac. He also introduced notation for naming these two types of vectors: a ket with name v is denoted by $|v\rangle$; a bra with name w is denoted by $\langle w|$. So we might write

$$|v\rangle = \begin{bmatrix} 2 \\ 0.5 \\ -3 \end{bmatrix} \text{ and } \langle w| = \begin{bmatrix} 1 & 0 & -\pi & 23 \end{bmatrix}.$$

Later we will see why we use two different symbols to surround the name, and the reason that tells us which side the angled bracket goes. But, for now, the important thing is to remember that kets refer to columns (think of the repeated "k" sound) and that bras, as usual, have their entries arranged horizontally.

Diagrams of Vectors

Vectors in two or three dimensions can be pictured as arrows. We will look at an example using $|a\rangle = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$. (In what follows we will often use kets for our examples, but if you like you can replace them with bras.) The first entry, 3 in this example, gives the change in the x-coordinate from the initial point to the terminal point. The second entry gives the change in the y-coordinate going from the initial point to terminal point. We can draw this vector with any initial point—if we choose (a, b) as the coordinates of its initial point, then the coordinates of its terminal point will be at $(a+3, b+1)$. Notice that if the initial point is drawn at the origin, the terminal point has coordinates given by the entries of the vector. This is convenient,

2 Linear Algebra

Quantum mechanics is based on linear algebra. The general theory uses infinite dimensional vector spaces. Fortunately for us, to describe spin or polarization we need only finite dimensions, which makes things much easier. In fact, we need only a few tools. At the end of this chapter I have given a list. The rest of the chapter explains how to use these tools and what the calculations mean. There are many examples. It is important to work carefully through all of them. The mathematics introduced here is essential to everything that follows. Like much mathematics, it can seem complicated when it is first introduced, but it becomes almost second nature with practice. The actual computations only involve addition and multiplication of numbers, along with an occasional square root and trigonometric function.

We will be using Paul Dirac's notation. Dirac was one of the founders of quantum mechanics, and his notation is used extensively throughout both quantum mechanics and quantum computing. It is not widely used outside these disciplines, which is surprising given how elegant and useful it is.

But first, we begin with a brief description of the numbers we will be using. These are real numbers—the standard decimal numbers with which we are all familiar. Practically every other book on quantum computation uses complex numbers—these involve the square root of negative one. So, let's start by explaining why we are not going to be using them.

Complex Numbers versus Real Numbers

Real numbers are straightforward to use. Complex numbers are—well, more complicated. To talk about these numbers we would have to talk about

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know that the photons coming through the first filter are polarized vertically. When measured by the second filter, half of the photons are found to be polarized in the 45° direction and half in the 135° directions. The ones with 45° polarization pass through the filter, and the others are absorbed. The third filter again measures the polarization in the vertical and horizontal directions. The photons entering have 45° polarization, and when measured in the vertical and horizontal directions, half will have vertical polarization and half will have horizontal polarization. The filter absorbs the vertically polarized photons and lets through those that are polarized horizontally.

Conclusions

We started this chapter by saying that classical bits can be represented by everyday objects like switches in the on or off position, but that qubits are generally represented by the spin of electrons or the polarization of photons. Spin and polarization are not nearly so familiar to us and have properties that are quite unlike their classical counterparts.

To measure spin, you first have to choose a direction and then measure it in that direction. Spin is *quantized*: When measured, it gives just two possible answers—not a continuous range of answers. We can assign classical bits to these results. For example, if we obtain an *N* we can consider it to be the binary digit 0, and if we obtain an *S* we can consider it to be the binary digit 1. This is exactly how we get answers from a quantum computation. The last stage of the computation is to take a measurement. The result will be one of two things, which will be interpreted as either 0 or 1. Although the actual computation will involve qubits, the final answer will be in terms of classical bits.

We have only just started our study, so we are quite limited in what we can do. We can, however, generate random strings of binary digits. The experiment that generated random strings of *N*s and *S*s can be rewritten as a string of 0s and 1s. Consequently measuring spins of electrons first in the vertical and then in the horizontal direction gives a random string of 0s and 1s. This is probably the simplest thing that we can do with qubits, but surprisingly this is something that cannot be done with a classical computer. Classical computers are deterministic. They can compute strings that pass various tests for randomness, but these are pseudorandom, not

then slide one of the squares over the other, there is minimal change. The amount of light let through the two overlapping squares is about the same as the amount that comes through each square when they are not overlapping. This is depicted in figure 1.8.

We will now rotate one of the squares through ninety degrees. Assuming we are not looking at light reflecting off a shiny surface, or light coming directly from a computer screen, but we are in normal light conditions, the proportion of horizontally polarized photons is equal to the proportion of vertically polarized ones, and both squares will look equally gray. We repeat the experiment of overlapping these squares. This time no light is let through the region of overlap, as depicted in figure 1.9.

The third experiment is to take the third sheet and rotate it through forty-five degrees. Under normal light conditions nothing appears to happen as we rotate the square. It maintains the same shade of gray. We now slide this square between the other two squares, one of which has vertical orientation, and the other has horizontal orientation. The result, as we noted earlier, is both surprising and unintuitive. Some light comes through the region of overlap of all three squares. (This is depicted in figure 1.10.) These polarized squares are sometimes called filters, but clearly they are

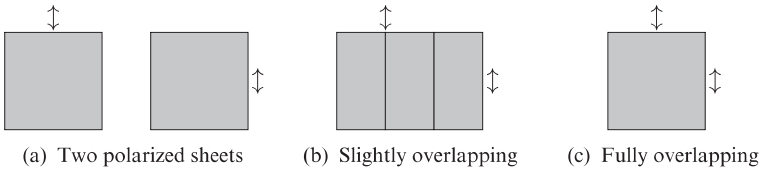


Figure 1.8
Two linear polarized squares with the same orientation.

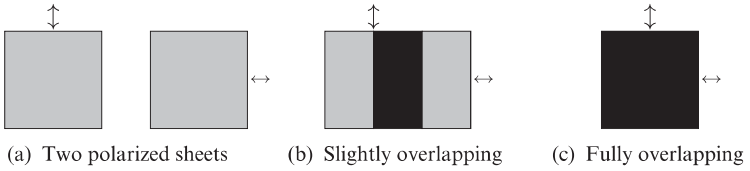


Figure 1.9
Two linear polarized squares with different orientations.

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each time we do it the initial conditions vary slightly. These slight variations can change the outcome from heads to tails and vice versa. There is no real randomness in classical mechanics, just what is often called *sensitive dependence to initial conditions*—a small change in the input can get amplified and produce an entirely different outcome. The underlying idea concerning randomness in quantum mechanics is different. The randomness is true randomness.

The sequence *NSSNNNSS* … that we obtained from measuring spin in two directions is considered to be truly random, as we shall see. The sequence of coin tosses, *HTTHHHTT* … appears random, but the classical laws of physics are deterministic and this apparent randomness would disappear if we could make our measurements with infinite accuracy.

At this stage it is natural to question this. Einstein certainly did not like this interpretation, famously saying that God does not play dice. Couldn't there be a deeper theory? If we knew more information about the initial configurations of our electrons, couldn't it be the case that the final results would no longer be random but completely determined? Couldn't there be *hidden variables*—once we know the values of these variables, the apparent randomness disappears? In what follows we will present the mathematical theory in which true randomness is used. Later we will return to these questions. We will describe a clever experiment to distinguish between the hidden variable and the true randomness hypotheses. This experiment has been performed several times. The outcomes have always shown that the randomness is real and that there is no simple hidden variable theory that can eliminate it.

We started this chapter by saying that a qubit can be represented by the spin of an electron or the polarization of a photon. We will show how the models for spin and polarization are related.

Photons and Polarization

It is often said that we are not aware of the strange quantum phenomena because they only occur at incredibly small scales and are not apparent at the scales of our everyday life. There is some truth to this, but there is an experiment that is completely analogous to measuring spin of electrons that can be performed with very little apparatus. It concerns polarized light.

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answers to the first and third questions need not be the same. For example, if we ask three times in a row if the hand is pointing toward twelve, we will get exactly the same answer each time, but if we ask first if it is pointing toward twelve, then whether it is pointing to three, and finally again whether it is pointing toward twelve, the answers to the first and third question need not be the same. The only difference between the two scenarios is the second question, so that question must be affecting the outcome of the following question. We will say a little more about these observations, starting with measurements.

Measurements

In classical mechanics, we might consider the path of a ball thrown into the air. The path can be calculated using calculus, but in order to perform the calculation we need to know certain quantities such as the mass of the ball and its initial velocity. How we measure these is not part of the theory. We just assume that they are known. The implicit assumption is that the act of measuring is not important to the problem—that taking a measurement does not affect the system being modeled. For the example of a ball being thrown into the air, this makes sense. We can measure its initial velocity using a radar gun, for example. This involves bouncing photons off the ball and, though bouncing photons will have an effect on the ball, it is negligible. This is the philosophy underlying classical mechanics: Measurements will affect the objects being studied, but experiments can be designed so the effect of measurement is negligible and so can consequently be ignored.

In quantum mechanics, we are often considering tiny particles like atoms or electrons. Here bouncing photons off them has an effect that is no longer negligible. In order to perform some measurement, we have to interact with the system. These interactions are going to perturb our system, so we can no longer ignore them. It should not seem surprising that measurement becomes a basic component of the theory, but what is surprising is how this is done. For example, consider the case where we measure the spin of an electron first in the vertical direction and then in the horizontal one. We have seen that exactly half of the electrons that have spin N in direction 0° after passing through the first detector will have spin N in direction 90° when measured by the second detector. It might seem that the strength of

Measurements in the Same Direction

Measurements are repeatable. If we repeat exactly the same measurement, we get exactly the same result. For example, suppose that we decide to measure an electron's spin in the vertical direction. We then repeat exactly the same experiment by positioning two more sets of our apparatus behind the first one. One is positioned in exactly the right place to catch electrons that are deflected upward by the first apparatus. The other is placed to catch the electrons deflected downward. The electrons that are deflected upward by the first apparatus are deflected up by the second, and the ones deflected down by the first apparatus are deflected down by the second. This means that electrons measured to have spin *N* in direction 0° initially also have spin *N* in direction 0° when we repeat the experiment. Similarly, if an electron is initially measured to have spin *S* in direction 0° and we repeat exactly the same experiment, it will still have spin *S* in direction 0°. For our clock analogy, if we repeatedly ask if the hand is pointing at twelve, we will repeatedly get the same answer: that either it is always pointing toward twelve or it is always pointing toward six.

There is, of course, nothing special about the vertical direction. If we start by measuring in direction $\theta°$, and then repeatedly measure in the same direction, we will obtain the exactly the same result each time. We will end up with a string of letters consisting entirely of *N*s or one entirely of *S*s.

The next thing to consider is what happens if we don't repeat the same measurement. For example, what happens if we first measure vertically and then horizontally?

Measurements in Different Directions

We will measure the electron's spin first in the vertical direction, then in the horizontal direction. We will send a stream of electrons through the first detector—measuring spin in the vertical direction. As before, we have two more detectors behind the first one in the appropriate positions to catch the electrons coming from the first detector. The difference is that these two detectors are rotated through 90° and measure spin in the horizontal direction.

First we look at the stream of electrons that are deflected upward by the first detector—these have spin *N* in direction 0°. When they go through the

from the twentieth century into an entirely new way of thinking about computation. The basic unit of quantum computing is the qubit. We will see what qubits are and what happens when we measure them. A classical bit is either 0 or 1. If it's 0 and we measure it, we get 0. If it's 1 and we measure 1, we get 1. In both cases the bit remains unchanged. The situation is totally different for qubits. A qubit can be in one of an infinite number of states—a superposition of both 0 and 1—but when we measure it, as in the classical case, we just get one of two values, either 0 or 1. The act of measurement changes the qubit. A simple mathematical model describes all of this precisely.

Qubits can also be entangled. When we make a measurement of one of them, it affects the state of the other. Again, this is something that we don't experience in our daily lives, but it is described perfectly by our mathematical model.

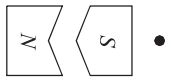
These three things—superposition, measurement, and entanglement—are the key quantum mechanical ideas. Once we know what they mean, we can see how they may be used in computations. It is here that human ingenuity enters the picture.

Mathematicians often describe proofs as being beautiful, often containing unexpected insights. I feel exactly the same way about many of the topics we will look at. Bell's theorem, quantum teleportation, superdense coding, all are gems. The error correcting circuit and Grover's algorithm are absolutely amazing.

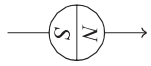
By the end of the book, you should understand the basic ideas that underlie quantum computing, and you will have seen some ingenious and beautiful constructions. You will also come to realize that quantum computing and classical computing are not two distinct disciplines, but that quantum computing is the more fundamental form of computing—anything that can be computed classically can be computed on a quantum computer. The qubit is the basic unit of computation, not the bit. Computation, in its essence, really means quantum computation.

Finally, it should be emphasized that this book is about the theory of quantum computation. It is about software, not hardware. We briefly mention hardware in places and occasionally talk about how to physically entangle qubits, but these topics are just asides. The book is not about how to build a quantum computer, but how to use one.

Here's a brief description of the book's contents.

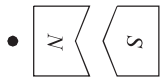


(a) Outcome of experiment

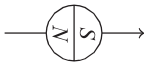


(b) Diagram of electron

Figure 1.4
Electron with spin N in the 90° direction.



(a) Outcome of experiment



(b) Diagram of electron

Figure 1.5
Electron with spin S in the 90° direction.

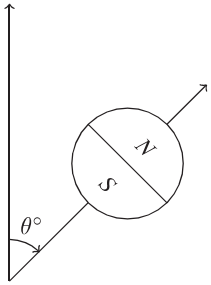


Figure 1.6
Electron with spin N in the θ° direction.

In the following chapters we will want to rotate the magnets through various angles. We will measure angles in the clockwise direction with 0° denoting the upward vertical direction and θ measuring the angle from the upward vertical. Figure 1.6 depicts an electron with spin *N* in the direction of a general angle θ°.

Sometimes spin is described as being *up*, *down*, *left*, or *right*. Our description of an electron being *N* in the direction 0° seems somewhat cumbersome, but it is unambiguous and avoids some of the pitfalls of using *up*, *down*, and so on, especially when we rotate the apparatus through 180°. For example, both of the situations pictured in figure 1.7 represent an electron having spin *N* in direction 0° or equivalently spin *S* in direction 180°.

Chapter 4. This chapter describes what it means for two qubits to be entangled. Entanglement is difficult to describe in words, but it is easy to describe mathematically. The new mathematical idea is the tensor product. This is the simplest way of combining mathematical models of individual qubits to give one model that describes a collection of qubits.

Though the mathematics is straightforward, entanglement is not something that we experience in everyday life. When one of a pair of entangled qubits is measured, it affects the second qubit. This is what Albert Einstein, who disliked it, called "spooky action at a distance." We look at several examples.

The chapter concludes by showing that we can't use entanglement to communicate faster than the speed of light.

Chapter 5. We look at Einstein's concerns with entanglement and whether a hidden variable theory can preserve local realism. We go through the mathematics of Bell's inequality—a remarkable result that gives an experimental way of determining whether or not Einstein's argument is correct. As most people know, Einstein's view was wrong, but even Bell thought he would be proved correct.

Artur Ekert realized that the setup for the test of Bell's inequality could also be used both to generate a secure key to be used for cryptography and at the same time to test whether any eavesdroppers are present. We conclude the chapter with a description of this cryptographic protocol.

Chapter 6. The chapter starts with standard topics in computation: bits, gates, and logic. Then we briefly look at reversible computation and the ideas of Ed Fredkin. We show that both the Fredkin gate and the Toffoli gate are universal—you can build a complete computer using only Fredkin gates (or Toffoli gates). The chapter concludes with Fredkin's billiard-ball computer. This is not really needed for the rest of our story, but its sheer ingenuity demands that it be included.

This computer consists of balls colliding with one another and off various walls. It conjures up images of particles interacting. This is one of the ideas that inspired Richard Feynman to become interested in the idea of quantum computing. Feynman wrote some of the earliest papers on the subject.

south magnet wins and the particle is deflected downward. After passing through the apparatus, the atoms are collected on a screen.

From the classical viewpoint, the magnetic poles of the atom could be aligned in any direction. If they were aligned horizontally, there would be no deflection, and in general, the size of the deflection would correspond to the amount the magnetic axis of the atom differs from the horizontal, with maximum deflections occurring when the magnetic poles of the atom are aligned vertically.

If the classical viewpoint is correct, when we send a large number of silver atoms through the machine we ought to see a continuous line on the screen going from the top point to the bottom. But this is not what Stern and Gerlach found. When they looked at the screen, they found just two dots: one at the extreme top and the other at the extreme bottom. All of the atoms behaved like little bar magnets that were aligned vertically. None of them had any other orientation. How could this be?

But before we start analyzing what is going on in more detail, we will shift our attention from atoms to electrons. Not only do atoms act like little magnets, but so also do their components. When we discuss quantum computers we will often talk about electrons and their spins. As with silver atoms, if you measure spin* in the vertical direction, you find that the electron is either deflected in the north direction or the south direction. Again, like silver atoms, you find that electrons are little magnets with their north and south poles perfectly aligned in the vertical direction. None of them have any other orientation.

In practice, you can't actually measure electron spin of a free electron using the Stern-Gerlach apparatus in the way we have shown because electrons have a negative charge and magnetic fields deflect moving charged particles. That said, the following diagrams give useful pictorial representations of the results of measuring spin in various directions. The idea behind this diagram is that you are the source; the magnets are lined up between you and this book. The dot shows how the electron gets deflected. In figure 1.2, the picture on the left shows the deflection by the magnets. The one on the right gives a depiction of the electron as a magnet with the north and south poles marked. We will describe this situation as saying the electron

* We will keep using the term *spin* because it is the standard terminology. But we are just determining the axis of the poles of a magnet.

solve but also for the new ideas they introduce. These underlying ideas have been and are being incorporated into a new generation of algorithms.

After looking at algorithms, we switch gears and briefly look at how quantum computation can be used to simulate quantum processes. Chemistry, at its most basic level, is quantum mechanical. Classical computational chemistry works by taking quantum mechanical equations and simulating them using classical computers. These simulations are approximations and ignore the fine details. This works well in many cases, but in some cases it doesn't. In these cases you need the fine details, and quantum computers should be able to give them.

This chapter also briefly looks at building actual machines. This is a very fast-growing area. The first machines are being offered for sale. There is even one machine available on the cloud that everyone can use for free. It looks likely that we will soon enter the age of *quantum supremacy*. (We explain what this means.)

The book concludes with the realization that quantum computation is not a new type of computation but is the discovery of the true nature of computation.

1 Spin

All computations involve inputting data, manipulating it according to certain rules, and then outputting the final answer. For classical computations, the *bit* is the basic unit of data. For quantum computations, this unit is the *quantum bit*—usually shortened to *qubit*.

A classical bit corresponds to one of two alternatives. Anything that can be in exactly one of two states can represent a bit. Later we will see various examples, which include the truth or falsity of a logical statement, a switch being in the on or off position, and even the presence or absence of a billiard ball.

A qubit, like a bit, includes these two alternatives, but—quite unlike a bit—it can also be in a combination of these two states. What does this mean? What exactly is a combination of two states, and what are physical objects that can represent qubits? What is the quantum computation analog to the switch?

A qubit can be represented by the spin of an electron or the polarization of a photon. This, though true, does not seem particularly helpful as spins of electrons and polarizations of photons are not things that most of us have knowledge about, let alone experience with. Let's start with a basic introduction to describe spin and polarization. To do this we describe the foundational experiment performed by Otto Stern and Walther Gerlach on the spin of silver atoms.

In 1922, Niels Bohr's planetary model described the current understanding of atoms. In this model an atom consisted of a positive nucleus orbited by negative electrons. These orbits were circular and were constrained to certain radii. The innermost orbit could contain at most two electrons. Once this was filled, electrons would start filling the next level, where at most eight electrons could be held. Silver atoms have 47 electrons. Two of